

SMART INDIA HACKATHON 2026



- **PS ID:** [verify SIH 2026 portal — previously observed SIH26157]
- **PS Title:** Supervisory Analytics Tool for SOC Assessment (SAT-SA)
- **Theme:** [verify portal — previously observed Blockchain & Cybersecurity]
- **Category:** Software [verify]
- **Team ID:** [actual submission ID]
- **Team Name:** [actual registered team name]





Proposed Solution (Describe your Idea/Solution/Prototype)

Periodic CSV, JSON and SQLite case submissions from **multiple CSEs** become a common object-centric event record.

Moat 1 joins explicit operational rules with structural anomalies and **execution gaps + negative space** peer baselines.

Each flagged issue opens an **explainable finding**: exact case, events, rule, rationale, provenance and review cost.

A budgeted review plan selects **diverse evidence** across capabilities instead of repeating the same easy cases.

Moat 2 asks whether a proposed intervention plausibly helps; uncertain estimates return to the **human examiner**.



Techstack Used:

- 1. Python 3.11+ — application and analytics
- 2. Streamlit — local examiner UI
- 3. DuckDB — SQL rule evaluation
- 4. OCEL 2.0 — in-house event/object model
- 5. SQLite — local batches and OCEL files
- 6. statsmodels + SciPy — peer/causal statistics
- 7. pgmpy — causal adjustment DAG
- 8. jsonschema + pytest — schema/test checks

Python

Streamlit

DuckDB

SQLite

OCEL 2.0

statsmodels

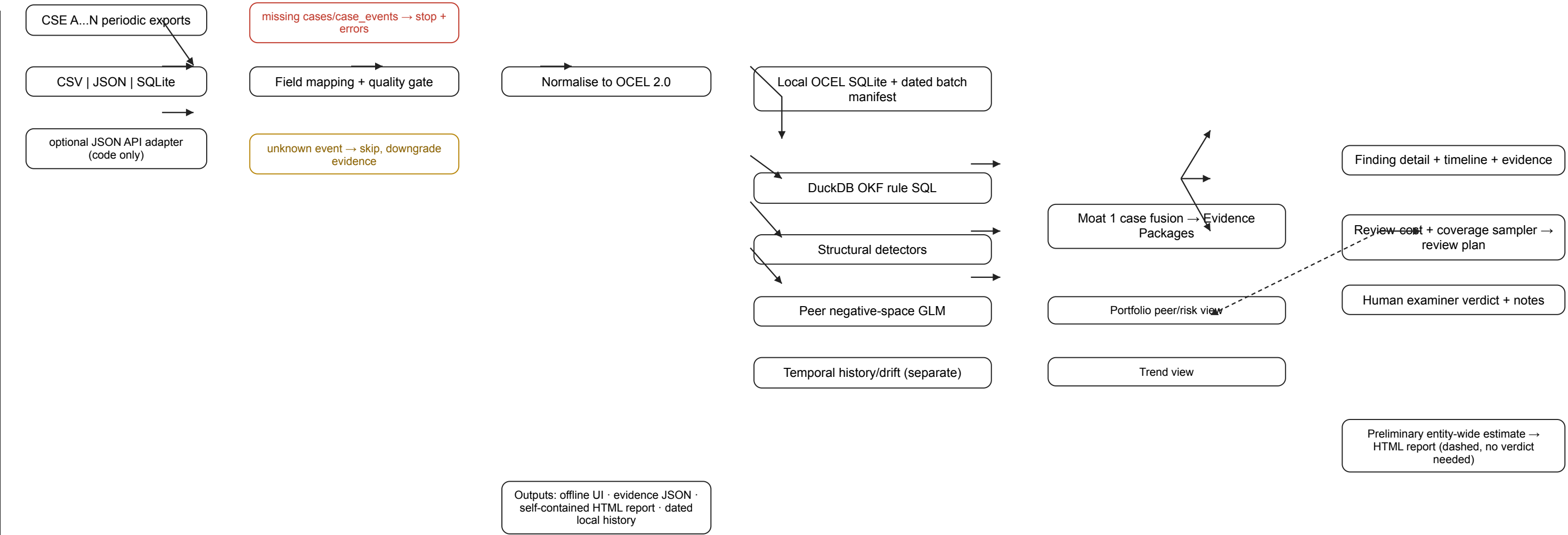
pgmpy

pytest

TECHNICAL APPROACH



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Prototype: [actual offline build/repository link]
Live offline prototype available for demonstration.

Feasibility dimension	Current evidence / workable route	Constraint and mitigation
Technical / offline	Python, Streamlit, DuckDB, SQLite and statistical packages run locally. CSV/JSON/SQLite imports, OCEL conversion, rules, findings, review plan, history and HTML/JSON exports are implemented.	Prepackage wheels, DuckDB SQLite extension and assets for a sealed network. Verify a clean-machine offline install. No cloud service or model API belongs in the runtime path.
Scale / accuracy	Multiple CSEs/batches work in prototype. Synthetic gates test OCEL integrity, planted patterns, fusion hard negatives, sampling diversity and causal abstention.	Current benchmarks are small synthetic profiles, not national-scale proof. Optimise ingestion/indexing and sampler for high finding counts; validate precision/recall against blinded manual reviews.
Data / governance	Minimum case and case-event exports support the core assessment; alerts/assets/analysts/queues enrich findings. Unknown event values are skipped with an evidence-quality downgrade.	Comparable asset/queue exposure, mapping quality and verified authority are needed for some peer/negative-space claims. Abstain or label provisional when evidence is inadequate.

Deployment and validation plan

- Air-gapped workstation/server; local processing and static reports.
- Back-test on expert-reviewed CSE submissions with per-type precision, recall and review-time measures.
- Scale-test by entity count, period count, event volume and concern count before operational release.

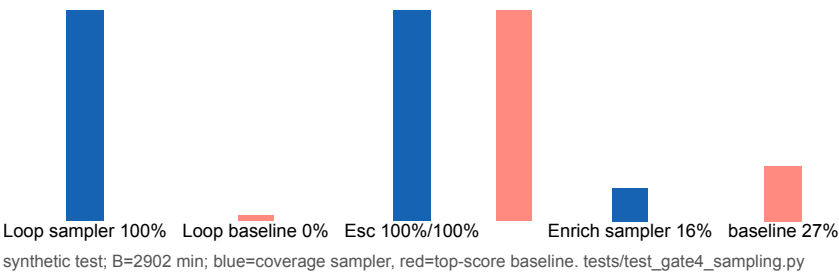
Supervisory impact

- 1. **Find where to look.** Rank CSEs and surface evidence gaps across comparable submissions.
- 2. **Inspect the full record.** Open a finding to see rule, affected case/asset, supporting events, timeline and evidence quality.
- 3. **Use review hours well.** Select varied findings under an examiner-set time budget.
- 4. **Keep judgement visible.** Record a verdict and notes; an uncertain intervention stays under investigation.

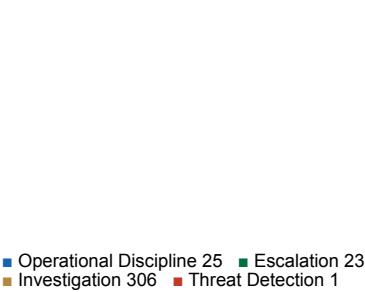
Practical benefits

- 1. **Across entities and periods.** Five synthetic demo CSEs plus imported dated batches, navigated in one local UI.
- 2. **Explainable by design.** Evidence JSON and standalone HTML reports expose the basis of each conclusion.
- 3. **Offline deployment fit.** No SaaS or externally hosted model is used by the implemented pipeline.
- 4. **Broader sampling.** Coverage-based selection reduces repeated review of near-duplicate finding types in tests.

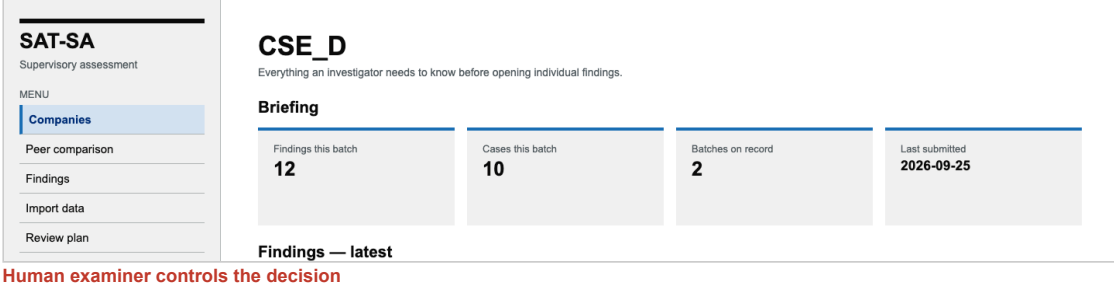
Per-type recall @ 30% budget (n=355, synthetic demo)



Findings by capability (n=355, same run)



Live examiner UI (Companies → batch → finding)





RESEARCH AND REFERENCES



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Research Survey

Prior approach / source	What it contributes	SAT-SA treatment and limit
Manual alert sampling and policy/KPI review	Expert judgement and rich context; costly and inconsistent at scale.	Evidence-first ranking and diverse sampling assist examiners; manual verdict remains final.
OCEL 2.0 standard	Models one event related to several objects.	In-house OCEL 2.0 model links cases, alerts, analysts, assets and queues.
Sviridenko (2004); Badanidiyuru–Vondrák (2014)	Diminishing-return coverage, budgeted selection.	Bounded-seed threshold-greedy heuristic; full formal guarantee does not transfer to the capped 12-item seed pool.
Poisson GLM (statsmodels)	Exposure-normalised expected counts and uncertainty.	Peer negative-space screen with abstention on inadequate evidence.
Cinelli–Hazlett sensitivity; pgmpy adjustment	Observational-effect sensitivity, backdoor identification.	Moat 2 reports a tentative effect; unmeasured confounding remains possible.

Reference

[1] OCEL Standard, “Object-Centric Event Log 2.0 Specification.” ocel-standard.org/2.0/ocel20_specification.pdf
[2] M. Sviridenko, “A note on maximizing a submodular set function subject to a knapsack constraint,” Operations Research Letters 32(1), 2004. [sciencedirect.com/.../S0167637703000622](https://www.sciencedirect.com/.../S0167637703000622)
[3] C. Cinelli and C. Hazlett, “Making Sense of Sensitivity: Extending Omitted Variable Bias,” JRSS B, 2020. doi.org/10.1111/rssb.12348

Dataset:
[Synthetic CSE A–E demo submissions](#) (generated/local sample inputs; not real NCIIPC data)
[OCEL/test fixtures and planted scenarios](#)
[Expert-reviewed field evaluation dataset](#): pending NCIIPC access and governance approval.